United International University

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AI Agents and the Future of Jobs

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Abstract

Artificial Intelligence (AI) is revolutionising the human working process. The latest and revolutionary advancement in this technology is the advent of AI agents. These independent systems can be given goals beyond the rules, can make decisions and can carry out a number of complex tasks with only a little human interaction.

AI agents are uniquely able to think multiple steps, plan and execute over multiple decisions and use digital tools to complete tasks. This report reviews how autonomous technology has systemically affected jobs around the world, evaluates the multi-faceted research challenge posed by operational, geographical and longitudinal dimensions of work, and suggests a viable methodological framework for human-technology partnerships to address the workforce challenge of autonomous technology.

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1 Introduction & Background

1.1 Evolution of Workplace Digitalization

Technology in the workplace has largely passed through four key phases in the last 20 years, as summarized in Table 1.

Table 1: Phases of Workplace Digitalization

Stage	**Type of Technology**	**Key Characteristics and Scope of Work**
Stage 1	Rule-Based Automation Software	Carries out simple and routine tasks in the office based on predetermined instructions (e.g., using Excel Macros).
Stage 2	Cognitive Software	Implements recommendations for humans using machine learning to interpret data and to make basic pattern recognition.
Stage 3	Generative Digital Assistant	Produces human text or images or code based on prompts or short instructions (e.g., Claude, ChatGPT).
Stage 4	Autonomous Cognitive Agents	Extremely intelligent computer programs that can operate in complex online environments—and may operate independently without the supervision of humans—to reach a complex result.

1.2 The Technological Shift in Labor Substitution

Past revolutions in automation, or "Industrial Revolutions," mostly touched manual and physical work of human labor; this is seen in the introduction of robots on factory assembly lines, for example. In contrast, self-autonomous AI agents are directly focused at projects requiring human thinking, administration, analysis and digital execution.

Today complex processes like software testing, legal document creation, market research and support to enterprise clients have been delivered by autonomous systems and thus, entire organizational pipelines managed throughout their life cycles without human interaction have been made possible.

2 Theoretical Paradigms in Labor Economics

Three major theories or paradigms can be observed in current economics regarding technological unemployment and workplace transformation.

2.1 The Task-Based Framework of Labor

A theory of the economy that assumes that any job can only be performed as a complex mixture of many small tasks. As a result of cognitive automation, a job basically falls into two categories:

1. **Routine Cognitive Tasks:** Intellectual tasks performed following the same rules, rules that are present every day (e.g., data entry, data filing, producing routine reports). Digital AI agents are likely to totally supersede these.
2. **Non-Routine Cognitive Tasks:** Tasks whose progress cannot be fully automated, like making strategic decisions, entering into complex negotiations, engaging in human creativity, and managerial leadership functions.

2.2 The Polarization of Skills in Knowledge Markets

The work place is becoming increasingly skill polarized because of the use of autonomous software:

- The market demand and compensation for high-demand managerial and administrative positions is increasing at a very rapid pace that include overseeing new technological frameworks, handling with AI governance, and designing policies.
- In contrast, the difficulty and earning power of entry-level, routine and lower intellectual work are rapidly decreasing. This imposes a lot of pressure on middle and lower middle/not-normal grade knowledge workers.

2.3 Macroeconomic Displacement Volatility

From a global economic perspective, two primary forces are at play in opposing directions:

- **Productivity Effect:** Improving production processes has helped to reduce the costs of producing products, enrich the world market and promote the development of a new industry.
- **Displacement Effect:** The potential loss of millions of knowledge workers from their jobs as knowledge jobs are displaced due to new automation technologies calls for immediate training to accommodate the new knowledge technologies.

3 Comprehensive Gap Analysis

While a great deal of research has been performed around the globe on how to automate the workforce, the following three areas we believe are under explored:

Gap 1: Operational Framework for Human-AI Collaboration (Operational Framework for HITL)

Most literature and research are concerned with two extremes; full automation (total replacement) or a normal human performing tasks with the support of technology.

- **Specific Gap:** The actual operational "Human-in-the-Loop" (HITL) framework for humans and autonomous agents is not concrete. In particular, research fails to address the following questions:
  - What will be the relative role of software and managers in ultimate decision making?
  - Who is accountable when there is a critical operational problem within an autonomous system?
  - If all the intermediate processes are carried out mechanically by the software, how can the degradation of the human intellect or capabilities be avoided with regard to the erosion?

Gap 2: Regional and Socio-Economic Impact – Low- and Middle-Income Countries

- **Issue:** Studies and data from empirical models drawn from existing studies are largely geared towards the G7 advanced countries (North America, Western Europe).
- **Specific Gap:** Research is totally ignored on the real socio-economic effect of such autonomous digital systems in a developing country as Bangladesh. However, the main effect of automation will be in countries such as Bangladesh and India where the locals have:
  - Large Business Process Outsourcing (BPO) industries and back office Customer Service (CS) hubs.
  - Entry to mid-level software development and IT export industry.
  - Millions of out-of-the-office freelancers are at a high risk of being eliminated by these new technologies.

Gap 3: Longitudinal Analysis – Compensation Effect vs. Net Displacement

- **Problem:** Study of the present is predominantly limited to short-term studies which only consider immediate work-speed or organizational effectiveness.

- **Specific Gap:** Due to the lack of long-term (5- to 10-year) data on the potential impacts of technology adoption on the number of jobs for the sector, papers were unable to assess whether the sector will

  1. experience a net loss of employment opportunities over time due to the "compensation effect" of lower prices and new markets created by technology adoption, or
  2. note an actual increase in employment.

4 Proposed Empirical Research Methodology

A combination of descriptive and qualitative methods was suggested for addressing the gaps identified:

- **Phase 1: Qualitative Interviews**

  Qualitative interviews with Enterprise CTOs, Heads of Operations and Heads of HR Directors in Software, Finance and Customer Service industries managing employees of 50 or more. The aim of this project is to clarify the boundaries of decision-making power between man and machine in real workplaces.

- **Phase 2: Quantitative Marketplace Analysis**

  Analyze long-term market data from the top-tier freelance marketplaces and enterprise management analytics platforms for accurate measurement of number of Junior/Senior roles, salary differences and project completion rates.

- **Phase 3: HITL Delegation Calculus**

  Design a mathematical model to calculate human to AI work, depending on different tasks complexity, risk cost, and system reliability:

$$DelegationLevel = f(TaskComplexity, OperationalRiskCost, SystemReliability) \quad (1)$$

5 Human-AI Integration Pipeline

The example of the pipeline below shows exactly how humans and AI agents will work together in the future work place and at what particular work tasks human oversight will remain.

Table 2: AI Agent Workflow Pipeline Steps

Step	**Pipeline Stage**	**Operational Action**	**Assigned Role**
1	Input & Objective Setting	Human provides project details, budget, & terms.	Human
2	Planning & Breakdown	AI agent analyzes goal & breaks it into sub-tasks.	AI Agent
3	Gateway & Approval Gate	Human reviews plan & approves or requests edits.	Human
4	Automated Execution	AI runs tasks automatically using web, code, or apps.	AI Agent
5	Quality Control & Audit	Human reviews output, fixes errors/hallucinations.	Human
6	Final Output & Delivery	Project delivered or released after final approval.	Human + AI

6 Detailed Pipeline Explanation

1. **Input and Objective Setting (Human):** Humans or managers feed the project's fundamental mission, budget, and timelines into the AI agent system.

2. **Planning and Breakdown (AI Agent):** The AI agent reviews the goal, segments it into sub-tasks and develops a strategy.

3. **Gateway and Approval (Human Approval Gate):** It is an important security step. The agent can move forward only when a human has looked at and greenlighted the plan developed by the AI agent.

4. **Automated Execution and Tool Usage (AI Agent):** When approved, the AI agent will automatically help them, search the web, execute code, or use Apps as needed to complete the task.

5. **Quality Control and Auditing (Human Quality Audit):** A human or quality auditor thoroughly reviews the final draft or code produced by the agent. Instructions are issued to rectify any errors or instances of "hallucination."

6. **Final Output and Delivery:** Following final human approval, the project is ready for delivery or market release.

7 Strategic Recommendations and Policy Roadmap

7.1 Policy and Governmental Action Plans

- Reform existing unemployment allowance policies to transition them into "transitional stipends" or "transition funds" specifically earmarked for workers' skill development.
- Establish specialized AI training programs at national and state-levels on data science, policy, ethics, AI governance, and AI management.
- Provide preferential tax and government incentives to companies investing in technology to augment workers and not to the point of eliminating job losses.

7.2 Enterprise and Organizational Strategies

- Avoid sudden job cutbacks due to cost-cutting in the business world, but instead put in place "human-centric" models of employment in which AI is used to assist human staff.
- Implement effective governance and audit mechanisms for data privacy and to ensure constant human control of automated processes. Ensure proper internal education and training to move general personnel to roles like process managers, quality auditors, and AI governance specialists.

7.3 Realignment of the Educational System

Eliminate the use of "analog" pedagogical approaches in schools and universities where only educational acts related to task execution or memorization are employed. Rather, re-design curricula to focus on high-level, managerial knowledge, critical thinking, system design, data architecture and governance of AI to equip students for the future job market.

8 Conclusion

Artificial Intelligence (AI) is creating another paradigm shift in this global economy. Though worries about the loss of jobs are quite justified, managing the transition period is not as easy as they think.

The success of AI agents for boosting productivity and promoting the human good will depend on overcoming the research challenges identified in this report, especially regarding the effects on developing countries such as Bangladesh and human-agent collaborative frameworks – HITL pipelines.

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